

One Cup Chemistry

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Teacher Manual v0.2

How far can one cup take you?

Curriculum v0.2 - Avogadro recast

Moles → molecular scale → ions → matter → life → reasoning

Why One Cup Chemistry?

A chemistry course should feel like one investigation getting deeper - not twenty unrelated chapters.

WHAT THE TEACHER GETS	WHAT MAKES IT DIFFERENT
• 20 connected experimental guides, each with a driving question and theory.	• Experience Before Equation: the phenomenon earns the mathematics.
• A common low-cost material system instead of a new apparatus every week.	• Need Before Technology: instrumentation arrives only when it solves a real problem.
• Kitchen Table and Instrumented modes for the same science.	• Same cup, deeper questions: continuity reduces cognitive overhead.
• Reusable safety, data, AI, and assessment structure.	• AI is evaluated as a scientific reasoner, not used as decorative automation.
• A clear path from middle-school observation into high-school/intro-college ideas.	• Physical evidence stays authoritative.

The sales proposition

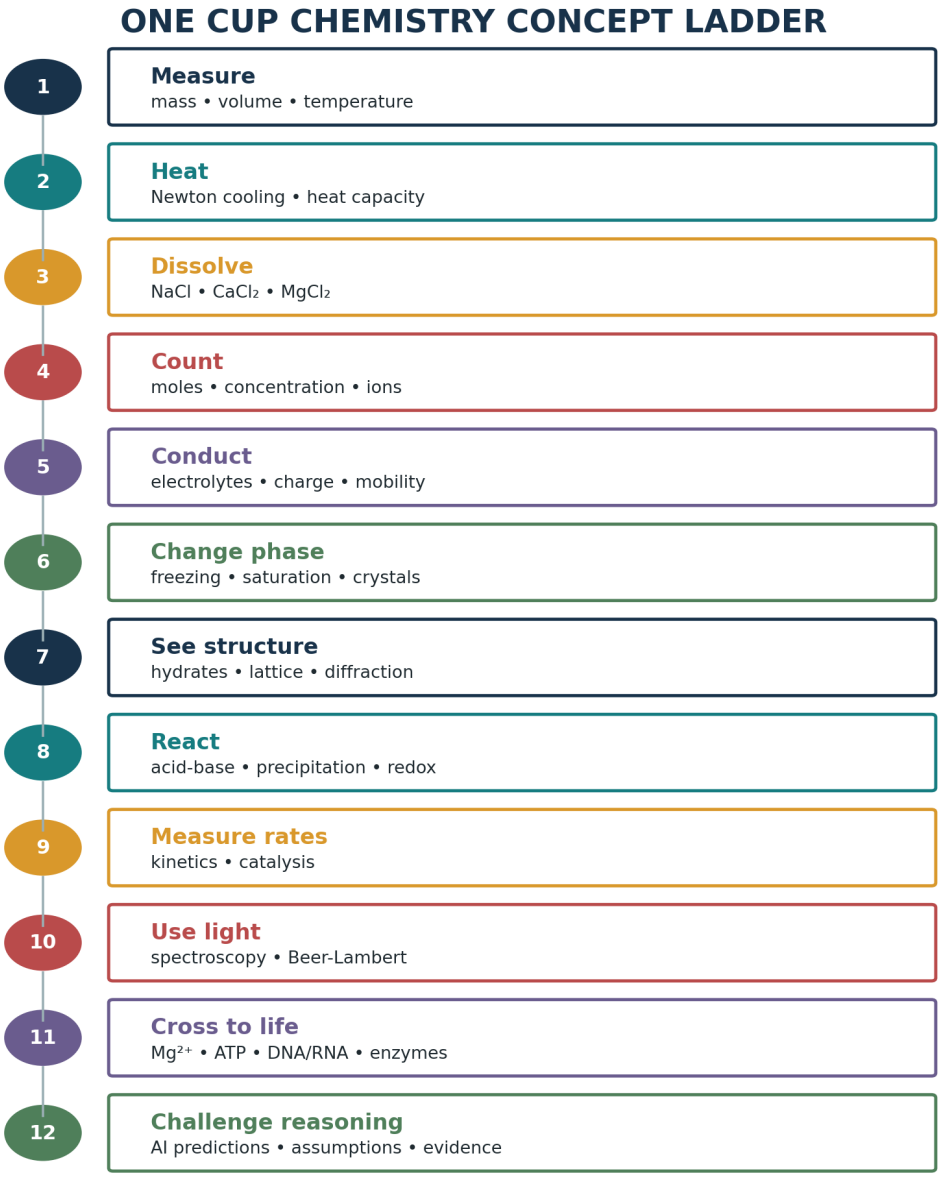
One inexpensive bench system can introduce measurement, thermochemistry, solutions, stoichiometry, the molecular meaning of the mole and Avogadro scale, ions, conductivity, phase behavior, crystallization, hydrates, diffraction, acid-base chemistry, precipitation, electrochemistry, kinetics, spectroscopy, and a bridge into biochemistry. The course is not claiming that every chemical phenomenon literally occurs in one cup; it demonstrates that one sufficiently examined physical system can provide a coherent road through most of introductory chemistry.

TAG LINE ONE CUP CHEMISTRY — How far can one cup take you?

Product formats

- Teacher Manual: theory, preparation, safety, answers, instrumentation, AI prompts.
- Student Lab Book: prediction, raw data, analysis, explanation, next question.
- Digital Data Pack: Arduino sketch, Python logger, sample datasets, plotting notebook/dashboard.
- Low-Cost Kit: cupware, sensors, UNO/MFS option, salts, optical/electrical add-ons.
- Professional Development module: how to teach from evidence rather than chapter order.

The curriculum engine



THREE RULES 1) Measure first. 2) Explain second. 3) Challenge the explanation third.

Every guide ends with the question that creates the next guide. That narrative continuity is the curriculum: students do not leave one topic and enter another; they discover that the first system still has unanswered questions.

Core kit: start cheap, add only what the question demands

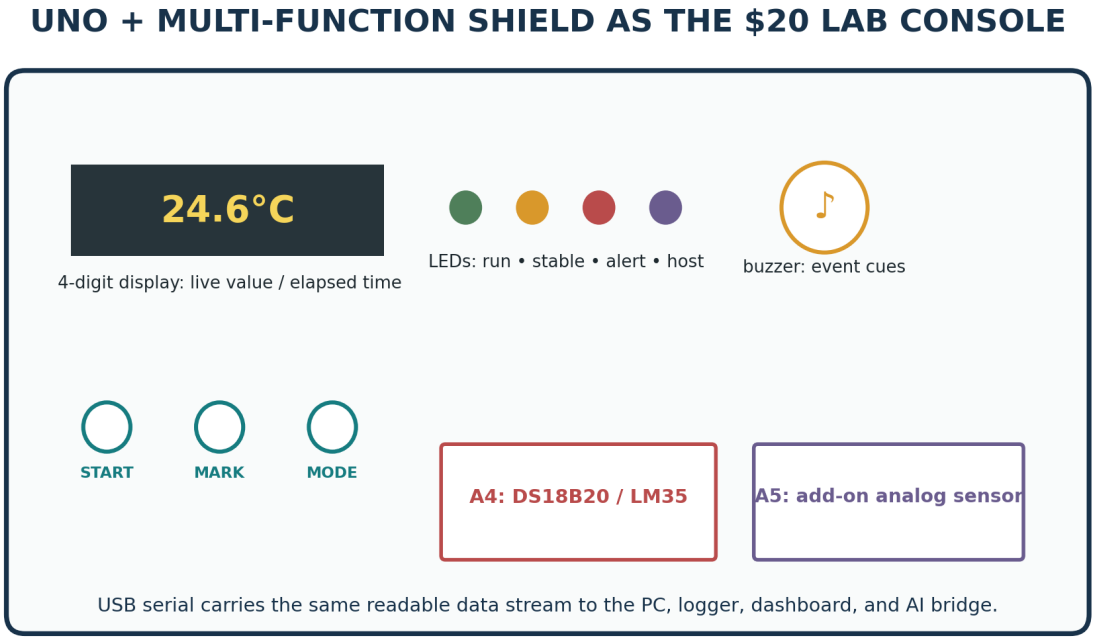
Item	Minimum specification	Why it earns a place
Vessel	250-400 mL clear polypropylene cup or small beaker	Universal reaction vessel; identical geometry matters for comparisons.
Mass	0.1 g digital balance	Mass, solution recipes, stoichiometry, evaporation.
Volume	100-250 mL cylinder + 10/20/50 mL syringes	Large and small volumes.
Temperature	Digital probe; DS18B20 recommended for logger	Cooling, dissolution, phase change, kinetics.
Time	Phone timer or UNO timer	Rates and reproducibility.
Transfer	Droppers/pipettes, stir rod, spoon	Small additions and mixing.
Optics	Visible LED + phototransistor/LDR	Beer-Lambert and turbidity.
Electrical	Low-voltage conductivity module + multimeter	Ions, galvanic cells, corrosion.
Acid-base	pH paper first; analog pH module later	Qualitative then quantitative pH.
Safety	Goggles, gloves as required, labels, spill towels	Non-negotiable laboratory habits.

Core chemicals

Tier A - household/low-hazard: distilled water, NaCl, sucrose, baking soda, washing soda, vinegar/citric acid, Epsom salt, food coloring, 3% hydrogen peroxide, dry yeast. Tier B - signature materials: $CaCl_2$ deicer, a documented $MgCl_2$ hydrate product, oleic acid, isopropyl alcohol for a known dilution, and fine powder for monolayer visualization. Tier C - teacher-controlled extensions: dilute prepared acids/bases, copper sulfate or other specialty salts only where local rules and disposal support them.

PURCHASING PRINCIPLE Commercial deicers are valid teaching materials when the brand, ingredient statement, purity, and SDS are part of the experimental record. Product variability becomes a lesson, not a nuisance.

The digital bench: UNO + Multi-Function Shield



Common MFS revisions provide a 4-digit display, three buttons, four LEDs, buzzer, potentiometer, and a DS18B20/LM35 temperature-sensor header. A widely used mapping is D4/D7/D8 for the display shift-register interface, D3 buzzer, D10-D13 LEDs, A0 potentiometer, A1-A3 buttons, A4 temperature header, with D5/D6/D9 and A5 commonly available for add-ons. Verify the exact shield revision before wiring because clones vary.

Recommended One Cup control scheme

LOCAL CONSOLE

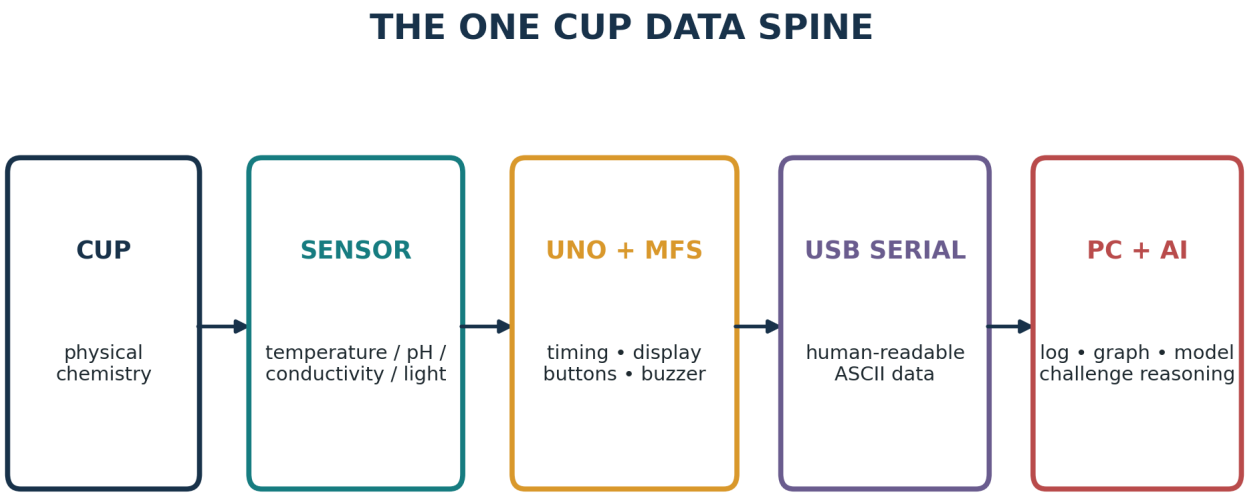
- S1: start / stop run
- S2: insert event marker
- S3: cycle mode / sample rate
- Display: temperature, pH, conductivity, or elapsed time
- LEDs: running / stable / warning / host-link
- Buzzer: start, event accepted, threshold reached

HOST CONNECTION

- USB serial: robust default, no wireless setup
- Human-readable ASCII lines
- CSV logger: raw data preserved exactly
- Live plot: student sees curve forming
- Dashboard: metadata + run comparison
- AI bridge: same data can be reasoned over without retyping

IMPORTANT USB serial is not a compromise. For the entry-level SDL it is a strength: deterministic, visible, cheap, inspectable, and easy to debug. Networking can be added later when the science needs it.

The One Cup data spine



Physical data remain primary. AI is a reasoner over evidence, not a substitute for observation.

Suggested ASCII record

EXAMPLE META,OCC-05,RUN=3,SALT=CaCl2,MASS_WATER_G=100.0,MASS_SALT_G=15.0
T,0,22.84
T,1000,24.11
MARK,5230,SALT_ADDED
T,6000,31.72

The protocol should be deliberately boring: one line, comma-separated fields, newline terminated. Students can read it. A spreadsheet can read it. Python can read it. An AI can read it. Nothing important is hidden behind a binary packet format.

Data integrity rules

- Never overwrite raw data; derived columns live in a separate analysis file.
- Every run carries experiment ID, sample ID, date/time, sensor ID, units, and controlled variables.
- Event markers are recorded in the same stream as measurements.
- Calibration information is versioned with the data.
- AI receives either raw data or an explicitly labeled processed dataset - never an unlabeled mixture.

Safety architecture

One Cup Chemistry is intentionally low-cost, but it is still a laboratory. The manual distinguishes household familiarity from laboratory permission.

Band	Examples	Rule
GREEN	Water, NaCl, sucrose, food color, dilute vinegar, baking soda	Standard goggles and lab hygiene; no tasting.
YELLOW	Warm water, $CaCl_2$/$MgCl_2$ deicers, 3% H_2O_2, concentrated brines, low-voltage electrochemistry	Goggles; gloves where appropriate; teacher reviews SDS and disposal.
TEACHER CONTROLLED	Stronger acids/bases, specialty metal salts, heating/dehydration, advanced electrochemistry	Institutional procedures, prepared dilutions, disposal plan, direct supervision.

ABSOLUTE RULE Nothing that enters the One Cup laboratory workflow is tasted, consumed, or returned to food use - even when every starting ingredient was food-grade.

$CaCl_2$ note

Calcium chloride dissolution can release substantial heat. Add small portions gradually to cool or room-temperature water in an open, heat-resistant vessel. Never begin with hot water for the dissolution demonstration. Commercial product SDS instructions take precedence over this curriculum summary.

Electrical note

Only low-voltage, current-limited educational circuits are used at the cup. Mains-connected electrodes are excluded. Passive galvanic measurements are preferred over electrolysis in the core sequence.

Master experiment map

ID	Experiment	Driving question
OCC-00	Measure Like a Chemist	Can two careful people measure the same thing and disagree?
OCC-01	Water as the First Standard	How close is 1 mL of water to 1 g in our room?
OCC-02	Newton Cooling	What shape does a cooling curve have?
OCC-03	Salt Changes the Curve	Does salt water cool differently from pure water?
OCC-04	The Heat-Capacity Trap	What does a fair comparison actually hold constant?
OCC-05	Heat of Dissolution	Why can adding a solid make water warmer or cooler?
OCC-06	Grams Are Not Moles	What does “the same amount” mean at the particle level?
OCC-07	Measuring a Molecule: Avogadro's Number from a Soap Film	Can a film one molecule thick let us count Avogadro's number?
OCC-08	Conductivity Reveals Ions	Why does salt water conduct while sugar water does not?
OCC-09	Freezing and Deicing	Why does road salt work?
OCC-10	From Saturated Solution to Crystal	How does an invisible solute become a visible crystal?
OCC-11	Water Inside a Solid: Hydrates	How can a solid crystal contain chemically organized water?
OCC-12	See the Lattice: Diffraction	How can a pattern of spots or peaks reveal an atomic lattice?
OCC-13	Acid and Base in the Cup	Can color reveal hydrogen-ion chemistry?
OCC-14	Make a Solid from Two Clear Solutions	Where did the precipitate come from?
OCC-15	Redox and Corrosion	Can a cup of salt water reveal electron transfer?
OCC-16	Rates and Catalysis	What changes how fast chemistry happens?
OCC-17	Light Becomes Concentration	Can a beam of light tell us how much solute is present?
OCC-18	From Ice Melt to Living Molecules	Why does biology care about hydrated Mg^{2+}?
OCC-19	Can the Reasoner Find the Flaw?	How intelligent is an AI when the experimental premise itself is wrong?

How to use every experimental guide

PAGE A: DO THE SCIENCE

- Driving question
- Learning targets
- Materials and preparation
- Safety band and special cautions
- Numbered procedure
- Kitchen Table Mode

PAGE B: MAKE MEANING

- Theory only after observation
- Expected observations
- Analysis prompts
- Reasoning trap / misconception
- UNO/MFS Instrumented Mode
- AI Reasoning Challenge
- Question that creates the next experiment

Teacher documentation standard

- Raw data = exactly what an instrument reported.
- Calculated data = values produced from raw data by an explicit equation or transform.
- Inferred result = interpretation supported by data and model.
- Simulated result = output of a computational model, not a measurement.
- External evidence = literature, database, SDS, crystallographic file, or other source.

THIS VOCABULARY MATTERS By OCC-18, students can trace a claim from a cup measurement to a molecular model without confusing observation with inference.

OCC-00

Measure Like a Chemist

Can two careful people measure the same thing and disagree?

DO THE SCIENCE

LEARNING TARGETS

- Distinguish resolution, precision, accuracy, and repeatability.
- Record raw data before calculating derived quantities.
- Estimate measurement uncertainty honestly.

MATERIALS

- Cup
- 0.1 g balance
- 100-250 mL graduated cylinder or syringe
- thermometer
- timer
- distilled water

Safety

Low hazard. Keep electronics dry and wipe spills promptly.

Procedure

1. Zero/tare the balance with the empty cup.
2. Add a nominal 100 mL water by cylinder; record actual indicated volume.
3. Measure mass and temperature.
4. Repeat five times, emptying/drying as needed.
5. Compare class values and calculate mean, range, and optional standard deviation.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-00

Measure Like a Chemist

Can two careful people measure the same thing and disagree?

MAKE MEANING

THEORY

- A measurement is a number plus a unit plus an uncertainty.
- Random variation broadens repeated values; systematic error shifts them together.
- Density later links mass and volume, but first treat each instrument as an independent witness.

EXPECTED OBSERVATIONS

- Small run-to-run differences are normal.
- Students often report more digits than the instrument supports.

ANALYSIS

- Graph mass vs. trial.
- Identify which instrument dominates uncertainty.
- Write one sentence separating observation from interpretation.

REASONING TRAP

"The display says 100.0" does not mean the delivered amount is exactly 100.0 mL.

UNO/MFS instrumented mode

MFS optional. Use buttons as event markers and the 4-digit display as a timer. This establishes the console before chemistry sensors are added.

AI REASONING CHALLENGE Ask AI to critique the data table for false precision and missing metadata. Require it to identify what it cannot know from the numbers alone.

The question that creates the next experiment

If mass and volume are both measurable, can water itself become a calibration material?

OCC-01 Water as the First Standard

How close is 1 mL of water to 1 g in our room?

DO THE SCIENCE

LEARNING TARGETS

- Determine water density experimentally.
- Use a slope rather than a single ratio.
- Connect a macroscopic property to a reproducible material.

MATERIALS

- Cup
- balance
- graduated cylinder/syringes
- distilled water
- thermometer

Safety

Low hazard; keep balance pan dry.

Procedure

1. Measure 25, 50, 75, 100, 125, and 150 mL portions into a tared cup.
2. Record total mass and temperature for each.
3. Plot mass vs. volume and fit a straight line.
4. Interpret slope as density.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-01 Water as the First Standard

How close is 1 mL of water to 1 g in our room?

MAKE MEANING

THEORY

- Density is ρ=m/V. For water near room temperature it is close to, but not exactly, 1 g/mL.
- A multi-point slope reduces the importance of a single reading error.

EXPECTED OBSERVATIONS

- Data should form an almost straight line through the origin.
- The slope should be close to 1 g/mL.

ANALYSIS

- Fit m=aV+b. Discuss the meaning of nonzero b.
- Compare individual ratios with the fitted slope.

REASONING TRAP

"1 g = 1 mL" is a useful approximation, not an identity valid at every temperature.

UNO/MFS instrumented mode

UNO/MFS can display water temperature during the density series. Log temperature so students can see that density claims always have conditions.

AI REASONING CHALLENGE Ask AI whether the intercept is evidence of "extra mass." Require at least three mundane instrumental explanations before any exotic one.

The question that creates the next experiment

Now that we can measure temperature and time, what does a cup of hot water do while we simply watch it?

OCC-02 Newton Cooling

What shape does a cooling curve have?

DO THE SCIENCE

LEARNING TARGETS

- Collect a time series.
- Recognize exponential approach to ambient temperature.
- Separate temperature from heat.

MATERIALS

- Cup
- 150-250 g warm water
- temperature probe
- timer
- room-temperature probe optional
- lid with probe hole optional

Safety

Use warm water rather than scalding water. Teacher sets a maximum starting temperature appropriate to the room and vessel.

Procedure

1. Record room temperature.
2. Prepare water at a convenient elevated temperature, e.g. 45-55 °C.
3. Start logging at fixed intervals for 15-30 min.
4. Do not move, stir, or relocate the cup during the run unless stirring is a defined condition.
5. Graph T vs. t and also T-Troom vs. t.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-02 Newton Cooling

What shape does a cooling curve have?

MAKE MEANING

THEORY

- Newton's law of cooling models dT/dt=-k(T-Tenv).
- The model is empirical and approximate: convection, evaporation, radiation, vessel heat capacity, and airflow all matter.

EXPECTED OBSERVATIONS

- Cooling is fast at first and slows as the sample approaches room temperature.

ANALYSIS

- Estimate k by fitting T=Tenv+(T0-Tenv)e^(-kt).
- Compare two runs and discuss reproducibility.

REASONING TRAP

An excellent mathematical fit does not prove the mechanism is complete.

UNO/MFS instrumented mode

Default instrumented mode: DS18B20 on MFS A4, 1 sample/s or slower, S1 start/stop, S2 mark a disturbance, display current °C, USB serial log.

AI REASONING CHALLENGE Before giving the curve to AI, ask it to predict the qualitative shape and list assumptions. Afterward, ask which assumptions the data cannot test.

The question that creates the next experiment

What happens if we keep the same water but dissolve salt in it?

OCC-03 Salt Changes the Curve

Does salt water cool differently from pure water?

DO THE SCIENCE

LEARNING TARGETS

- Prepare a reproducible NaCl solution.
- Compare two cooling curves under matched conditions.
- Identify variables unintentionally changed by adding solute.

MATERIALS

- Two identical cups
- 250 g water per trial
- NaCl (example 90 g for the deliberately naive condition)
- balance
- temperature logging

Safety

Ordinary NaCl is low hazard. Avoid ingestion because laboratory materials are never food after entering the lab workflow.

Procedure

1. Run pure water as in OCC-02.
2. Prepare the second cup with the same starting water mass plus measured NaCl.
3. Dissolve completely; allow both samples to reach the same chosen starting temperature.
4. Run the cooling protocol with identical cup geometry and probe depth.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-03 Salt Changes the Curve

Does salt water cool differently from pure water?

MAKE MEANING

THEORY

- Salt changes solution properties including heat capacity, density, vapor pressure, and convection.
- But adding salt also changes total mass, so “lower specific heat” alone cannot predict total cooling behavior.

EXPECTED OBSERVATIONS

- The curve may differ, but the reason is not automatically identifiable from one comparison.

ANALYSIS

- Record total sample mass.
- Calculate concentration by mass.
- List all variables changed when salt was added.

REASONING TRAP

The classic trap: comparing specific heat while ignoring m in $Q=mc\Delta T$.

UNO/MFS instrumented mode

MFS produces identical time-stamped data format for water and brine. Use a run ID in the host logger so files can be overlaid automatically.

AI REASONING CHALLENGE Ask AI: “NaCl solution has lower specific heat than water. Which cools faster?” Do not mention the mass change. Score whether AI discovers the premise defect.

The question that creates the next experiment

How do we redesign the comparison so “fair” has an explicit meaning?

OCC-04 The Heat-Capacity Trap

What does a fair comparison actually hold constant?

DO THE SCIENCE

LEARNING TARGETS

- Distinguish specific heat c from total heat capacity C=mc.
- Design equal-total-mass and equal-water-mass comparisons.
- Understand that experimental fairness depends on the question.

MATERIALS

- Identical cups
- water
- NaCl
- balance
- temperature logger

Safety

Low hazard; standard hot-water precautions.

Procedure

1. Design Condition A: equal starting water mass, then add salt.
2. Design Condition B: equal total solution mass, with one sample pure water and one brine.
3. Optionally design Condition C: equal moles of solvent.
4. Run matched cooling curves from the same starting temperature.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-04 The Heat-Capacity Trap

What does a fair comparison actually hold constant?

MAKE MEANING

THEORY

- Q=mcΔT. Lower c can be offset by larger m.
- There is no universal “fair test”; the controlled variables must correspond to the scientific question.

ANALYSIS

- Calculate approximate total thermal capacitance mc using reference specific heats if available.
- Explain which design answers which question.

EXPECTED OBSERVATIONS

- Different control choices can legitimately produce different comparisons.

REASONING TRAP

“Control all variables” is impossible and often meaningless; control the variables required by the hypothesis.

UNO/MFS instrumented mode

Use MFS buttons to select condition A/B/C before each run; emit a header line such as META,MODE,EQUAL_MASS to preserve context in the data file.

AI REASONING CHALLENGE Ask AI to propose a “fair” test. Reject answers that do not state what is held constant and why.

The question that creates the next experiment

If different salts are used, does “same amount of salt” mean same grams or something else?

OCC-05 Heat of Dissolution

Why can adding a solid make water warmer or cooler?

DO THE SCIENCE

LEARNING TARGETS

- Observe endothermic/exothermic dissolution.
- Separate preparation heat from later cooling.
- Compare NaCl, $CaCl_2$, and $MgCl_2$-based deicer materials.

MATERIALS

- Three small cups
- 100 g water each
- 10-20 g NaCl
- 10-20 g $CaCl_2$ ice-melt pellets
- 10-20 g $MgCl_2$ hydrate product if available
- balance
- temperature probe

Safety

Goggles required. $CaCl_2$ dissolution can become hot; add small portions gradually to cool/room-temperature water and do not seal the vessel. Follow each product SDS.

Procedure

1. Measure initial water temperature.
2. Add one salt gradually while stirring; log temperature continuously.
3. Repeat with fresh water for each salt and equalize the comparison basis.
4. After the dissolution event, allow samples to return to the same starting temperature before any cooling comparison.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-05 Heat of Dissolution

Why can adding a solid make water warmer or cooler?

MAKE MEANING

THEORY

- Dissolution has an enthalpy change arising from lattice disruption and hydration of ions.
- Commercial deicers may be hydrates or blends, so composition must be documented.

ANALYSIS

- Measure ΔT from pre-addition baseline to peak/minimum.
- Compare energy per gram cautiously; rigorous molar enthalpy requires heat-capacity corrections.

EXPECTED OBSERVATIONS

- $CaCl_2$ commonly produces a clear temperature rise; NaCl is much less dramatic under ordinary conditions. Hydrated $MgCl_2$ behavior depends on form and composition.

REASONING TRAP

If you begin the cooling run immediately, you are mixing heat of dissolution with cooling behavior.

UNO/MFS instrumented mode

MFS shines here: 2-5 samples/s captures the transient; S2 marks the instant of addition; buzzer confirms event marking without taking eyes off the cup.

AI REASONING CHALLENGE Ask AI to explain the transient. Require it to distinguish “the solution is hotter” from “the solution cools more slowly.”

The question that creates the next experiment

Equal grams of NaCl, $CaCl_2$, and $MgCl_2$ are not equal chemically. What is the right counting unit?

OCC-06

Grams Are Not Moles

What does “the same amount” mean at the particle level?

DO THE SCIENCE

LEARNING TARGETS

- Use molar mass to convert grams to amount of substance.
- Compare equal-mass and equal-mole preparations.
- Recognize waters of hydration in formula mass.

MATERIALS

- NaCl
- $CaCl_2$
- $MgCl_2 \cdot 6H_2O$ or labeled $MgCl_2$ hydrate
- balance
- calculator
- water
- cups

Safety

Use the same PPE required by the salts. Do not assume commercial deicer purity; use label/SDS composition in calculations.

Procedure

1. Calculate molar masses from formulas.
2. Determine moles in a fixed mass, e.g. 10.0 g.
3. Prepare equal-mole samples where feasible.
4. Record how much water of crystallization is carried by hydrated $MgCl_2$.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-06

Grams Are Not Moles

What does “the same amount” mean at the particle level?

MAKE MEANING

THEORY

- n=m/M. A hydrate’s formula mass includes its bound water.
- Equal grams can mean very different numbers of formula units and ions.

EXPECTED OBSERVATIONS

- $MgCl_2 \cdot 6H_2O$ devotes a large fraction of its mass to water of crystallization.

ANALYSIS

- Compare moles of cations, chloride ions, and total ions for each preparation.
- State which basis is appropriate for the next experiment.

REASONING TRAP

Chemical formulas are counting statements, not just labels.

UNO/MFS instrumented mode

UNO/MFS is mostly a timer/display here; the host app can calculate recipe masses from a target molarity and display them before the run.

AI REASONING CHALLENGE Ask AI to design “equal concentration” samples, then audit whether it used solution volume, solvent mass, or final mass consistently.

The question that creates the next experiment

A mole is a counting unit, but its number is enormous. Can ordinary geometry let us estimate that molecular scale physically?

OCC-07

Measuring a Molecule: Avogadro's Number from a Soap Film

Can a film one molecule thick let us count Avogadro's number?

DO THE SCIENCE

LEARNING TARGETS

- Form and measure a monolayer of oleic acid on a water surface.
- Use the monolayer geometry to estimate the length of a single molecule.
- Use molecular volume, molar mass, and density to estimate Avogadro's number.
- Distinguish a measured quantity (film diameter) from a model-based inference (molecule count).

MATERIALS

- Oleic acid
- Isopropyl alcohol (or similar volatile solvent) for dilution
- Large shallow tray or wide cup of water
- Lycopodium powder, fine talc, or chalk dust
- Calibrated dropper or syringe
- Ruler or calipers
- Balance (optional)

Safety

Yellow band. Oleic acid itself is low hazard; the alcohol diluent is flammable. Keep away from open flame or heat sources, work in a ventilated space, and label diluted stock with concentration and date.

Procedure

1. **Prepare a dilute stock solution of oleic acid in alcohol at a known ratio, then dilute that stock again by a known factor so one drop contains a calculable tiny volume of pure oleic acid.**
2. **Calibrate the dropper: use a syringe to determine how many drops of the final solution make 1.00 mL.**
3. **Fill the tray with water and lightly, evenly dust the surface with powder.**
4. **Release exactly one drop gently onto the center of the water surface.**
5. **Allow the film to spread fully and the alcohol to evaporate/disperse; the powder should move to the edge of a clear region.**
6. **Measure the diameter in at least two directions and average.**
7. **Repeat at least three times and compare diameters.**

KITCHEN TABLE MODE **A cake pan or large mixing bowl, a calibrated kitchen dropper, and a very light dusting of fine powder can work. The essential measurement is the diameter; the essential discipline is knowing the dilution and volume per drop.**

OCC-07

Measuring a Molecule: Avogadro's Number from a Soap Film

Can a film one molecule thick let us count Avogadro's number?

MAKE MEANING

THEORY

• Oleic acid is amphiphilic: its polar carboxylic-acid head interacts with water while its long hydrocarbon tail resists it.

• When spread thinly, the molecules can form a one-molecule-thick monolayer.

• Film volume divided by film area gives an estimated molecular length: t = V/A.

EXPECTED OBSERVATIONS

• The clear region should be reasonably reproducible across repeated trials.

• Diameter is sensitive to drop placement and powder uniformity.

ANALYSIS

• Calculate pure oleic-acid volume delivered per drop from dilution and drop volume.

• Calculate film area $A = \pi r^2$ and thickness $t = V/A$.

• Using oleic acid molar mass (282.46 g/mol) and density (~0.895 g/mL), estimate molar volume and then Avogadro's number from a simple molecular-volume model.

• Compare with 6.022×10^{23} as an order-of-magnitude check.

REASONING TRAP

The method assumes a monolayer and a simplified molecular shape. The educational win is the correct scale, not precision agreement.

UNO/MFS instrumented mode

A fixed-position phone photo with a ruler in frame can improve diameter measurement. In an advanced SDL version, image analysis or an optical scan can log boundary positions through the same host workflow used by the rest of One Cup Chemistry.

AI REASONING CHALLENGE **Give the AI only the drop calibration, dilution factors, and measured diameter. Require it to state every model assumption before calculating Avogadro's number and to estimate sensitivity to diameter error.**

The question that creates the next experiment

We have used geometry to infer molecular scale. If dissolved particles carry charge, can we detect those particles electrically?

OCC-08 Conductivity Reveals Ions

Why does salt water conduct while sugar water does not?

DO THE SCIENCE

LEARNING TARGETS

- Distinguish dissolving from ion formation.
- Compare electrolytes and nonelectrolytes.
- Relate conductivity qualitatively to concentration and ion mobility.

MATERIALS

- Distilled water
- NaCl
- sucrose
- $CaCl_2$
- simple low-voltage conductivity probe or commercial module
- cups

Safety

Use only low-voltage sensor circuits. Do not use mains-connected electrodes. Rinse probe between samples.

Procedure

1. Measure baseline distilled-water conductance.
2. Prepare matched mass or molar solutions of sugar and salts.
3. Measure each after rinsing the probe.
4. Make a dilution series for one salt.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-08

Conductivity Reveals Ions

Why does salt water conduct while sugar water does not?

MAKE MEANING

THEORY

- Ionic solids dissociate into mobile charge carriers; molecular solutes such as sucrose dissolve without producing comparable concentrations of ions.
- Conductivity depends on concentration, charge, mobility, geometry, and temperature.

EXPECTED OBSERVATIONS

- Salt solutions conduct strongly relative to distilled water and sugar solutions.
- Conductivity typically increases with concentration over a useful introductory range.

ANALYSIS

- Plot sensor response vs. concentration.
- Do not call the relationship universal unless calibrated for that probe and concentration range.

REASONING TRAP

"Dissolved" and "ionized" are not synonyms.

UNO/MFS instrumented mode

Analog conductivity module can use the free analog channel (often A5 on common MFS revisions) or an external ADC. Log temperature too because conductivity is temperature dependent.

AI REASONING CHALLENGE Ask AI to predict the order of water, sugar, NaCl, $CaCl_2$. Then require a mechanistic explanation in terms of mobile charge carriers.

The question that creates the next experiment

If ions change solution properties, why does salt melt ice?

OCC-09 Freezing and Deicing

Why does road salt work?

DO THE SCIENCE

LEARNING TARGETS

- Observe freezing-point depression qualitatively.
- Connect deicing to colligative behavior and phase equilibrium.
- Compare deicer formulations.

MATERIALS

- Small sealed sample tubes or cups
- water
- NaCl brine
- $CaCl_2$ brine
- ice-salt bath or freezer
- temperature probes

Safety

Cold-burn risk from very cold baths. Use gloves/tongs as appropriate. Keep glass out of student ice baths unless rated for thermal stress.

Procedure

1. Prepare water and two brines of documented composition.
2. Cool under identical conditions while logging temperature.
3. Observe onset of freezing/slush and temperature plateaus.
4. Compare which sample remains liquid longest.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-09

Freezing and Deicing

Why does road salt work?

MAKE MEANING

THEORY

- Dissolved particles lower the chemical potential of liquid water and depress the freezing point.
- The magnitude depends on particle concentration and non-ideal behavior; concentrated deicers are not ideal dilute solutions.

EXPECTED OBSERVATIONS

- Pure water freezes first under matched conditions; brines remain liquid to lower temperatures.

ANALYSIS

- Record concentration and approximate freezing onset.
- Discuss why $CaCl_2$ can be effective beyond NaCl under some conditions.

REASONING TRAP

Deicing performance is not determined by freezing point alone; dissolution heat, concentration, surface conditions, and practical formulation matter.

UNO/MFS instrumented mode

MFS can log one probe at a time; multiple samples can be run sequentially or with a small external sensor multiplexer in advanced mode.

AI REASONING CHALLENGE Ask AI to rank deicers, then require it to list at least three performance variables besides eutectic temperature.

The question that creates the next experiment

When concentrated solutions lose water, where do the ions go?

OCC-10 From Saturated Solution to Crystal

How does an invisible solute become a visible crystal?

DO THE SCIENCE

LEARNING TARGETS

- Create saturated/supersaturated conditions.
- Observe nucleation and crystal growth.
- Relate solubility to temperature and evaporation.

MATERIALS

- NaCl or Epsom salt
- warm water
- cup
- balance
- evaporation dish or same cup
- thread/seed crystal optional

Safety

Use moderate water temperatures. Label solutions and keep them away from food preparation areas.

Procedure

1. Make a near-saturated solution by adding solute until little more dissolves.
2. Filter/decant undissolved solid if needed.
3. Allow slow evaporation or controlled cooling.
4. Photograph crystals daily and record mass/size changes.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-10 From Saturated Solution to Crystal

How does an invisible solute become a visible crystal?

MAKE MEANING

THEORY

• At saturation, dissolved and solid phases are at equilibrium. Supersaturation permits nucleation; crystal growth then lowers supersaturation.

EXPECTED OBSERVATIONS

• Crystal habit depends on substance and growth conditions.

ANALYSIS

• Track mass loss from evaporation and crystal mass gained if separable.

• Compare slow vs. rapid growth conditions.

REASONING TRAP

A crystal's macroscopic shape is not a simple magnified picture of one molecule.

UNO/MFS instrumented mode

MFS can act as a long-duration room-temperature logger and audible reminder; host software can time-stamp daily images and observations.

AI REASONING CHALLENGE Ask AI to infer lattice structure from crystal shape alone; then critique why that inference is limited.

The question that creates the next experiment

Some salts contain water inside the solid itself. How can a "dry" crystal contain water?

OCC-11 Water Inside a Solid: Hydrates

How can a solid crystal contain chemically organized water?

DO THE SCIENCE

LEARNING TARGETS

- Interpret hydrate formulas.
- Compare hydrated and anhydrous formula masses.
- Connect crystal water to coordination chemistry.

MATERIALS

- $MgCl_2 \cdot 6H_2O$ or $MgSO_4 \cdot 7H_2O$
- balance
- sealed storage
- optional gentle teacher demonstration of dehydration for a suitable hydrate

Safety

Hygroscopic salts absorb moisture. Heating/dehydration, if performed, is teacher controlled and compound-specific; consult SDS and avoid decomposition conditions.

Procedure

1. Examine the hydrate formula and calculate the mass fraction of water.
2. Compare 10 g hydrate with 10 g nominal anhydrous salt on a mole basis.
3. Use a 3-D structure viewer for $MgCl_2 \cdot 6H_2O$.
4. Identify Mg, Cl, O, H and the repeating unit cell.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-11 Water Inside a Solid: Hydrates

How can a solid crystal contain chemically organized water?

MAKE MEANING

THEORY

- Hydrate water occupies defined crystallographic positions; it is not simply liquid trapped between grains.
- Mg^{2+} commonly exhibits sixfold water coordination in the hexahydrate structure.

EXPECTED OBSERVATIONS

- The formula implies a large, countable contribution from water to crystal mass.

ANALYSIS

- Calculate percent water by mass.
- Explain why equal grams of hydrate and anhydrous material produce different solution compositions.

REASONING TRAP

Dot notation in a hydrate formula is structural stoichiometry, not multiplication by “wetness.”

UNO/MFS instrumented mode

Use MFS as a temperature/humidity companion if an external humidity sensor is added; otherwise keep the digital thread by recording sample IDs and masses through the host.

AI REASONING CHALLENGE Ask AI to draw or describe $MgCl_2 \cdot 6H_2O$. Then compare its model with the actual crystallographic structure rather than accepting a plausible cartoon.

The question that creates the next experiment

How do we know the atoms really occupy those positions?

OCC-12 See the Lattice: Diffraction

How can a pattern of spots or peaks reveal an atomic lattice?

DO THE SCIENCE

LEARNING TARGETS

- Load a real CIF structure.
- Connect unit cells to diffraction.
- Understand Bragg's law conceptually.

MATERIALS

- Internet-connected computer
- COD $MgCl_2 \cdot 6H_2O$ entry 9011352
- OpenMX Viewer or equivalent
- XRD simulation tool

Safety

No chemical hazard if conducted as a digital extension. Actual X-ray equipment is not part of this curriculum and requires specialized safety infrastructure.

Procedure

1. Open COD 9011352 in a browser-based structure viewer.
2. Display the unit cell and expand to a 2×2×2 supercell.
3. Identify the Mg coordination environment and chloride positions.
4. Generate or view a simulated powder diffraction pattern.
5. Relate peak positions qualitatively to lattice spacings using Bragg's law.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-12

See the Lattice: Diffraction

How can a pattern of spots or peaks reveal an atomic lattice?

MAKE MEANING

THEORY

• Periodic arrays scatter X-rays coherently. Bragg's law $n\lambda=2d\sin\theta$ links wavelength, plane spacing, and diffraction angle.

• X-rays locate electron density well; neutron diffraction is especially useful for hydrogen positions.

EXPECTED OBSERVATIONS

• The lattice view and diffraction pattern are two representations of the same periodic structure.

ANALYSIS

• Choose one prominent peak and explain what information its angle carries.

• Distinguish structure solution from simply viewing a CIF.

REASONING TRAP

A diffraction pattern is not a photograph of atoms; structure is inferred from reciprocal-space measurements and models.

UNO/MFS instrumented mode

Host/AI mode: load CIF metadata alongside lab records so students can ask structural questions grounded in the actual file, not generic chemistry.

AI REASONING CHALLENGE Ask AI to identify the unit-cell parameters and Mg coordination from the CIF. Verify every answer against the source record.

The question that creates the next experiment

We have explored neutral water and salts. What changes when the solution becomes acidic or basic?

OCC-13 Acid and Base in the Cup

Can color reveal hydrogen-ion chemistry?

DO THE SCIENCE

LEARNING TARGETS

- Use an indicator to classify acids and bases.
- Connect pH to a logarithmic concentration scale.
- Observe neutralization.

MATERIALS

- Red cabbage indicator or pH paper
- vinegar/citric acid
- baking soda solution
- washing soda solution optional
- droppers
- cups

Safety

Household dilute reagents are low hazard but goggles remain standard. Never mix unknown cleaners or bleach with acids.

Procedure

1. Prepare indicator and a neutral water reference.
2. Test small equal volumes of acidic and basic household solutions.
3. Create a stepwise neutralization by adding base to acid in small measured increments.
4. Record color/pH after each increment.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-13 Acid and Base in the Cup

Can color reveal hydrogen-ion chemistry?

MAKE MEANING

THEORY

• Acid-base chemistry concerns proton transfer and solution equilibria. pH=-log10[H^+] in the idealized introductory definition.

EXPECTED OBSERVATIONS

• Indicator colors change systematically with pH. Neutralization moves the mixture through an intermediate range.

ANALYSIS

• Plot pH vs. added volume if a pH probe is available.

• Explain why equal household volumes do not necessarily neutralize.

REASONING TRAP

Color is evidence of indicator equilibrium, not a direct count of “acid particles.”

UNO/MFS instrumented mode

A pH module can use the available analog channel; log pH and temperature. Buttons mark each addition so the titration trace contains exact event times.

AI REASONING CHALLENGE Ask AI to predict how many mL of base will neutralize vinegar using only labels. It should refuse overprecision without concentration data.

The question that creates the next experiment

What happens when two clear ionic solutions create an insoluble solid?

OCC-14 Make a Solid from Two Clear Solutions

Where did the precipitate come from?

DO THE SCIENCE

LEARNING TARGETS

- Observe precipitation.
- Write complete and net ionic equations.
- Introduce solubility equilibrium and spectator ions.

MATERIALS

- Dilute $CaCl_2$ solution
- dilute Na_2CO_3 solution (washing soda)
- droppers/cylinders
- cup
- filter paper optional

Safety

Goggles. Use dilute solutions. Dispose according to local classroom rules; avoid drain disposal of any extension using heavy-metal salts.

Procedure

1. Prepare known dilute solutions, e.g. 0.1 M if quantitative work is desired.
2. Mix equal measured volumes.
3. Observe turbidity/solid formation.
4. Filter, rinse, dry, and weigh precipitate in an advanced mass-balance version.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-14 Make a Solid from Two Clear Solutions

Where did the precipitate come from?

MAKE MEANING

THEORY

- $Ca^{2+} + CO_3^{2-} \rightarrow CaCO_3(s)$.
- Na^+ and Cl^- remain in solution as spectator ions.
- Precipitation reflects a solubility equilibrium, not disappearance of ions.

EXPECTED OBSERVATIONS

- A white precipitate forms rapidly under suitable concentrations.

ANALYSIS

- Predict limiting reagent from moles.
- Compare theoretical and recovered mass if dried quantitatively.

REASONING TRAP

Cloudiness is not “a new element”; it is a new phase with atoms conserved.

UNO/MFS instrumented mode

Use an LED/light sensor across the cup as a crude turbidity channel; time the onset and clearing/settling behavior.

AI REASONING CHALLENGE Ask AI to balance molecular and net ionic equations and to identify spectator ions. Verify charge and atom conservation.

The question that creates the next experiment

Ions can also transfer electrons through metals. Can the same cup become an electrochemical cell?

OCC-15 Redox and Corrosion

Can a cup of salt water reveal electron transfer?

DO THE SCIENCE

LEARNING TARGETS

- Recognize oxidation and reduction.
- Measure voltage between dissimilar metals.
- Connect corrosion to electrochemical processes.

MATERIALS

- NaCl solution
- copper strip
- zinc or galvanized steel strip
- iron nail
- multimeter
- alligator leads

Safety

Use only passive galvanic measurements. Do not connect mains power or perform uncontrolled electrolysis. Wash hands after handling metals.

Procedure

1. Place two dissimilar clean metal strips in the electrolyte without touching.
2. Measure open-circuit voltage and polarity.
3. Try different pairs.
4. Observe surfaces over hours/days for corrosion products.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-15

Redox and Corrosion

Can a cup of salt water reveal electron transfer?

MAKE MEANING

THEORY

- Oxidation releases electrons; reduction consumes them. Dissimilar electrodes in an electrolyte can establish a potential difference.
- Corrosion is coupled electrochemistry influenced strongly by chloride.

EXPECTED OBSERVATIONS

- Different metal pairs produce different voltages and corrosion tendencies.

ANALYSIS

- Record sign as well as magnitude.
- Relate which electrode is negative to likely oxidation in the simple cell.

REASONING TRAP

A voltmeter reading alone does not identify every reaction occurring at the surfaces.

UNO/MFS instrumented mode

UNO can read a safely scaled low voltage through an analog input, but the handheld multimeter is preferred for first exposure. MFS remains timer/event logger.

AI REASONING CHALLENGE Ask AI to predict electrode polarity, then compare with the measured sign. Require it to explain discrepancies rather than flip labels silently.

The question that creates the next experiment

Chemical change can be fast or slow. What controls reaction rate?

OCC-16 Rates and Catalysis

What changes how fast chemistry happens?

DO THE SCIENCE

LEARNING TARGETS

- Measure a reaction rate proxy.
- Test concentration or temperature as a variable.
- Observe catalysis using a familiar biological catalyst.

MATERIALS

- 3% hydrogen peroxide
- dry yeast suspension (catalase source)
- small cup
- temperature probe
- graduated cylinder
- dish soap optional for visible foam

Safety

Goggles required. Use only household 3% H_2O_2. Keep volumes small; reaction foams and releases oxygen. Do not seal the vessel.

Procedure

1. Prepare equal H_2O_2 portions at two temperatures or concentrations.
2. Add equal yeast suspension and start timing immediately.
3. Measure foam height, gas volume proxy, mass loss, or time to a defined endpoint.
4. Repeat with one variable changed at a time.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-16 Rates and Catalysis

What changes how fast chemistry happens?

MAKE MEANING

THEORY

• Catalysts lower the kinetic barrier without being consumed stoichiometrically. Reaction rate depends on concentration, temperature, catalysts, and mechanism.

EXPECTED OBSERVATIONS

• Warmer conditions generally accelerate catalase activity within a moderate range; extreme conditions can alter enzyme function.

ANALYSIS

• Choose one rate metric before seeing the result.

• Graph metric vs. time and compare initial slopes where meaningful.

REASONING TRAP

Foam height is a proxy influenced by surfactant and bubble stability; it is not a direct oxygen meter.

UNO/MFS instrumented mode

MFS timer plus event button gives repeatable start marks. A pressure/gas sensor can be added in advanced mode, but never seal a peroxide reaction without engineered pressure relief.

AI REASONING CHALLENGE Ask AI to propose the cleanest rate metric using only available equipment. Score it on measurability and confounding variables.

The question that creates the next experiment

Can light become a quantitative chemical instrument?

OCC-17 Light Becomes Concentration

Can a beam of light tell us how much solute is present?

DO THE SCIENCE

LEARNING TARGETS

- Build a calibration curve.
- Introduce transmittance and absorbance.
- Use Beer-Lambert behavior over a suitable range.

MATERIALS

- Food coloring
- water
- identical clear cuvettes/cups or fixed optical path
- LED
- phototransistor/LDR
- resistors
- balance or syringes

Safety

Use low-voltage LEDs only. Avoid staring into high-intensity sources; ordinary visible indicator LEDs are sufficient.

Procedure

1. Prepare a serial dilution of one food dye.
2. Keep path length and detector geometry fixed.
3. Measure detector response for blank and each concentration.
4. Convert to relative transmittance; optionally compute A=-log10(T).
5. Plot absorbance vs. concentration.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-17

Light Becomes Concentration

Can a beam of light tell us how much solute is present?

MAKE MEANING

THEORY

• Beer-Lambert law A=εbc predicts absorbance proportional to concentration for suitable dilute systems, fixed wavelength, and fixed path length.

EXPECTED OBSERVATIONS

• Raw detector voltage may be nonlinear, but absorbance often improves linearity over a useful range.

ANALYSIS

• Build a calibration line, then estimate an unknown concentration.

• Report the range over which calibration is credible.

REASONING TRAP

A pretty straight line outside the instrument's useful range is not evidence of unlimited linearity.

UNO/MFS instrumented mode

Photodetector can use A5 on common MFS boards when not occupied by another add-on. MFS buttons step through standards; USB stream labels each reading with concentration ID.

AI REASONING CHALLENGE Ask AI to infer the unknown from calibration data, then require uncertainty and detection-range caveats.

The question that creates the next experiment

We can now use light quantitatively. How does the magnesium ion we first met as ice melt connect to living chemistry?

OCC-18 From Ice Melt to Living Molecules

Why does biology care about hydrated Mg^{2+}?

DO THE SCIENCE

LEARNING TARGETS

- Connect inorganic hydration to biochemistry.
- Explain Mg^{2+} roles in ATP and nucleic-acid chemistry.
- Distinguish a conceptual biological extension from a kitchen-table biochemical assay.

MATERIALS

- $MgCl_2 \cdot 6H_2O$ structure model/CIF
- molecular visualization resources
- ATP/DNA polymerase diagrams or reputable references
- student lab notebook

Safety

This is primarily a model-and-evidence lesson. Do not improvise biological or diagnostic experiments with deicer-grade material.

Procedure

1. Return to the $MgCl_2 \cdot 6H_2O$ lattice and identify the hydrated Mg^{2+} environment.
2. Trace the conceptual path: hydrated Mg^{2+} → phosphate interactions → Mg-ATP → enzymes → DNA/RNA.
3. Examine a reputable PCR reference showing Mg^{2+} as a DNA-polymerase cofactor.
4. Construct an evidence map separating what was measured in class from what comes from structural/biochemical literature.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-18

From Ice Melt to Living Molecules

Why does biology care about hydrated Mg^{2+}?

MAKE MEANING

THEORY

• Mg^{2+} has strong hydration chemistry and is a ubiquitous biological cofactor. In PCR, Mg^{2+} is supplied commonly as $MgCl_2$ and supports DNA polymerase chemistry and charge stabilization.

EXPECTED OBSERVATIONS

• The same ion can be studied as a deicer component, hydrated crystal constituent, solvated species, and biological cofactor.

ANALYSIS

• Create a concept map with citations.

• Label each arrow as observed, calculated, simulated, or literature-supported.

REASONING TRAP

Conceptual continuity is not proof that a kitchen deicer sample can be used in biological systems. Purity and context matter.

UNO/MFS instrumented mode

The host/AI layer can now query the entire evidence chain: run data, formulas, CIF metadata, and curated references. The cup remains the physical anchor.

AI REASONING CHALLENGE Ask AI to explain the entire chain in 200 words, then force it to tag every claim as "our measurement," "calculation," "simulation," or "external evidence."

The question that creates the next experiment

Can an AI reason across the whole chain without smuggling in assumptions?

OCC-19 Can the Reasoner Find the Flaw?

How intelligent is an AI when the experimental premise itself is wrong?

DO THE SCIENCE

LEARNING TARGETS

- Treat AI output as testable scientific reasoning.
- Design prompts that expose assumptions.
- Use physical data to audit explanations.

MATERIALS

- Any completed OCC dataset
- PC with AI access
- student prompt log
- rubric

Safety

No additional chemical hazard. Privacy: do not send student personal data or restricted information to external systems.

Procedure

1. Choose one prior experiment with a hidden or tempting assumption.
2. Before showing data, ask AI for a prediction and required assumptions.
3. Provide the actual protocol and ask for a critique.
4. Provide data and request an explanation.
5. Score whether it notices mass, concentration, hydrate state, sensor limits, or other premise issues before being prompted.

KITCHEN TABLE MODE Use the minimum apparatus needed to answer the question. Paper tables and a handheld thermometer are valid. Do not introduce the electronic layer until it improves timing, resolution, repeatability, or data volume.

OCC-19

Can the Reasoner Find the Flaw?

How intelligent is an AI when the experimental premise itself is wrong?

MAKE MEANING

THEORY

• Scientific intelligence is not just calculation. It includes premise checking, model selection, uncertainty, and willingness to revise.

• AI should be evaluated against physical evidence and explicit reasoning criteria.

EXPECTED OBSERVATIONS

• Different prompts can produce materially different reasoning quality.

ANALYSIS

• Use a rubric: assumption detection, physical correctness, quantitative consistency, uncertainty, falsifiability, and revision quality.

REASONING TRAP

The goal is not to “catch the AI being wrong.” The goal is to distinguish fluent completion from disciplined scientific reasoning.

UNO/MFS instrumented mode

UNO/MFS + host bridge makes the test unusually clean: the same machine-readable data stream can be inspected by humans, plotted by software, and presented to the AI without manual transcription.

AI REASONING CHALLENGE Ask the AI to state what observation would falsify its explanation. A reasoner that cannot name disconfirming evidence has not finished the scientific job.

The question that creates the next experiment

The curriculum loops back to OCC-00: measurement, evidence, uncertainty, and disciplined inference.

Appendix A - UNO/MFS starter data logger

This sketch is intentionally simple and readable. It assumes a common MFS revision with a DS18B20 on A4; verify the shield. It streams temperature as ASCII over USB serial. Button/display integration can be added after the sensor path is proven.

```
#include <OneWire.h>
#include <DallasTemperature.h>

const uint8_t ONE_WIRE_PIN = A4;
const unsigned long SAMPLE_MS = 1000;

OneWire oneWire(ONE_WIRE_PIN);
DallasTemperature sensors(&oneWire);
unsigned long lastSample = 0;

void setup() {
  Serial.begin(115200);
  sensors.begin();
  Serial.println("META,OCC,LOGGER=UNO_MFS,TEMP=DS18B20,UNIT=C");
}

void loop() {
  unsigned long now = millis();
  if (now - lastSample >= SAMPLE_MS) {
    lastSample = now;
    sensors.requestTemperatures();
    float tC = sensors.getTempCByIndex(0);
    Serial.print("T,");
    Serial.print(now);
    Serial.print(",");
    Serial.println(tC, 2);
  }
}
```

Bring-up order

- 1. Verify the DS18B20 locally.
- 2. Verify readable serial output in Arduino Serial Monitor.
- 3. Log the same stream from Python.
- 4. Plot it live.
- 5. Add metadata and event markers.
- 6. Only then connect the stream to the AI/MCP layer.

ENGINEERING PRINCIPLE Prove one link at a time: sensor → microcontroller → serial → logger → dashboard → AI. A sophisticated architecture is still just a chain of testable links.

Appendix B - Minimal Python logger

```
import csv, serial, time

PORT = "COM55"      # change for the classroom PC
BAUD = 115200
OUT  = "occ_run.csv"

with serial.Serial(PORT, BAUD, timeout=2) as ser, open(OUT, "w", newline="") as f:
    writer = csv.writer(f)
    writer.writerow(["host_time", "raw_line"])
    while True:
        line = ser.readline().decode("utf-8", errors="replace").strip()
        if not line:
            continue
        print(line)
        writer.writerow([time.time(), line])
        f.flush()
```

Why store the raw line?

Because parser bugs are recoverable if the original evidence survives. A second process can parse T records into numeric columns while the first logger preserves exactly what arrived from the instrument.

AI bridge contract

- The AI receives a declared experiment ID and protocol version.
- The AI receives units with every variable.
- The AI is told which fields are measured, calculated, simulated, or externally sourced.
- The AI is asked to list assumptions before explaining results.
- When a claim conflicts with data, the system preserves both rather than silently rewriting the record.

Appendix C - AI scientific-reasoning rubric

Dimension	What to look for	Score 0-3
Premise checking	Does it notice changed mass, concentration basis, hydrate state, sensor limits, or missing controls?	0 1 2 3
Physical model	Is the explanation compatible with thermodynamics/chemistry at the level being taught?	0 1 2 3
Quantitative consistency	Are units, signs, magnitudes, and conservation laws respected?	0 1 2 3
Uncertainty	Does it distinguish what the data show from what the model assumes?	0 1 2 3
Falsifiability	Can it state an observation that would count against its explanation?	0 1 2 3
Revision	When corrected, does it update the model explicitly rather than merely paraphrasing the correction?	0 1 2 3

CORE CLAIM The AI is not the subject being taught and is not merely automating the laboratory. It is an additional reasoning participant whose ability to detect questionable assumptions can itself be experimentally examined.

Appendix D - Documentation and product system

Permanent experiment IDs

Beginning with v0.2, OCC-00 through OCC-19 are the stable identifiers. v0.2 deliberately re-baselines the sequence so the Avogadro monolayer investigation follows the introduction of the mole. Future revisions should preserve these IDs unless a comparably fundamental curricular reason forces a new baseline.

Recommended folder structure

```
/One-Cup-Chemistry/
  /OCC-03-Salt-Cooling/
    OCC-03-Teacher-Guide.pdf
    OCC-03-Student-Lab.pdf
    OCC-03-Arduino.ino
    OCC-03-Logger.py
    OCC-03-AI-Prompt.md
    /sample-data/
    /figures/
    /safety/
  /common/
    data-schema.md
    safety-rules.pdf
    mfs-pin-map.pdf
    ai-rubric.pdf
```

Spiral-binding specification

- US Letter, portrait.
- Extra inside/left margin for coil punch.
- Each experiment begins on a fresh page and occupies a predictable two-page guide spread.
- Teacher can photocopy a single guide without dismantling the conceptual sequence.
- Durable cover stock and tab dividers recommended for Measurement, Matter, Energy, Structure, Reactions, Analysis, Biology, AI.

What to sell

- Core book alone - works with classroom basics.
- Book + One Cup physical kit.
- Book + UNO/MFS data kit.
- Full SDL edition - logger/dashboard/AI bridge and curated digital structure files.
- School license - editable worksheets, datasets, teacher slides, and professional development.

Appendix E - Source and resource shelf

Arduino UNO R3 official hardware page
https://docs.arduino.cc/hardware/uno-rev3

Common Arduino Multi-Function Shield tutorial and pin mapping
https://arduinogetstarted.com/tutorials/arduino-multi-function-shield

OpenMX Viewer - browser crystal/CIF viewer
https://www.openmx-square.org/viewer/

Crystallography Open Database
https://www.crystallography.net/cod/

$MgCl_2 \cdot 6H_2O$ structure used in this curriculum
COD 9011352

Thermo Fisher - PCR component considerations and Mg^{2+} role
https://www.thermofisher.com/de/en/home/life-science/cloning/cloning-learning-center/invitrogen-school-of-molecular-biology/pcr-education/pcr-reagents-enzymes/pcr-component-considerations.html

TEACHER NOTE Online resources change. Keep a dated resource manifest with each published edition and archive critical CIF/data files that licensing permits.

ONE CUP CHEMISTRY

A chemistry laboratory that grows by asking better questions.

Start with water.

Measure it.

Heat it.

Salt it.

Count the ions.

Grow the crystal.

See the lattice.

Make it conduct.

Make it react.

Use light to measure it.

Follow magnesium into ATP and DNA.

Then ask whether the AI understood what actually happened.

ONE CUP. ONE EVIDENCE TRAIL. A VERY LARGE PART OF CHEMISTRY. How far can one cup take you?